



SIMATS ENGINEERING

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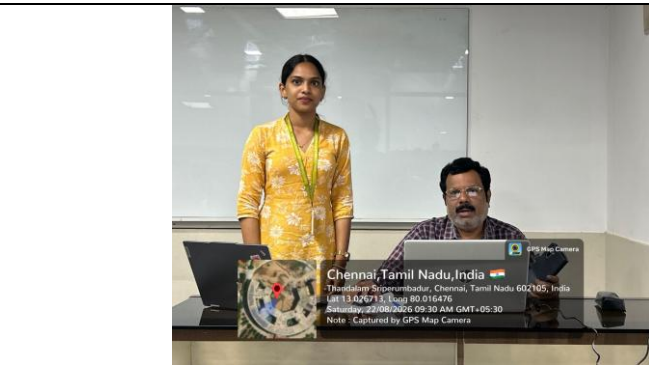
Course Code:ITA 0406

Slot:C

Course Name: R PROGRAMMING

Course Faculty: Dr.M.Amanullah

**Project Title:** Data Cleaning, Exploratory Analysis and Predictive Modelling of Farm Sensor Data Using R



**Project Description:** (here you write what you did in this project (contribution) including Model Description

In this project, I collected and prepared farm sensor data to analyze agricultural conditions and predict crop yield using R Programming. My contribution included working with important parameters such as soil moisture, temperature, rainfall, crop type, and yield, and organizing the sensor data systematically. I prepared the dataset by identifying missing observations, detecting invalid sensor readings, and correcting inconsistent values using suitable data-cleaning techniques. I also transformed the data into a structured format and calculated useful features such as average soil moisture, average temperature, and total rainfall for analysis. The cleaned dataset was then processed in R Studio, where it was used for exploratory data analysis, visualization, and predictive modelling. Through this process, I developed a reliable and structured dataset that could be used to understand relationships between farm conditions and crop yield.

*Reshni Priya*

Student Signature

Guide Signature